



A
Project Report
on
VIRTUAL FASHION ADVISOR:
AN AI-BASED OCCASION-ORIENTED DRESS
RECOMMENDATION SYSTEM
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May, 2026

DECLARATION

We hereby declare that the project report entitled “**Virtual Fashion Advisor: An AI-Based Occasion-Oriented Dress Recommendation System**” submitted for the award of the degree of Bachelor of Technology is our original work. The work presented in this report has been carried out by us under the supervision of the faculty guide.

To the best of our knowledge, the work presented in this report has not been submitted previously for the award of any other degree or diploma of this or any other institution. Proper acknowledgment has been given wherever reference has been made to the work of others.

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ABSTRACT

Fashion recommendation is an emerging research field which connects technology to daily life. People often find it difficult to select attire which is appropriate for a particular occasion, comfortable, as well as socially or culturally acceptable. The Virtual Fashion Advisor (VFA) proposed in this paper attempts to solve this issue by recommending clothes and draping styles for men and women. Traditional approaches for fashion recommendation systems have aimed largely at e-commerce personalization. This system attempts to emphasize on occasion specific appropriateness and cultural correctness.

The initial prototype employs Python dictionaries for rule-based mapping of attire with filters (occasion, time of day, comfort, and gender). The architecture is designed for scalability into ML-driven models, with integration possibilities for real-time datasets and virtual try-on systems. Testing with a sample dataset yielded an estimated accuracy of $\sim 82\%$ in matching user expectations with recommendations. This paper discusses the system architecture, methodology, dataset schema, implementation details, limitations, and future enhancements to make the VFA, a comprehensive digital fashion companion.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
VFA	Virtual Fashion Advisor
ML	Machine Learning
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
API	Application Programming Interface
NLP	Natural Language Processing
CV	Computer Vision
UI	User Interface
AI	Artificial Intelligence
UX	User Experience
JSON	JavaScript Object Notation
KNN	K-Nearest Neighbors
GNN	Graph Neural Network
AR	Augmented Reality
DL	Deep Learning

CHAPTER 1

INTRODUCTION

1.1 Introduction

Fashion is an essential part of modern life and acts as a medium for self-expression, cultural representation, and professional presentation.

The concept of a Virtual Fashion Advisor aims to bridge this gap by providing outfit recommendations that are tailored to specific occasions and user preferences. The system analyzes factors such as event type, gender, time of day, and comfort level to recommend appropriate clothing combinations. This approach helps users avoid inappropriate fashion choices and reduces decision fatigue when selecting outfits.

The proposed system uses a rule-based recommendation engine implemented in Python. The rule engine maps user inputs to outfit suggestions using predefined fashion guidelines. Unlike complex machine learning systems that require large datasets and training processes, the rule-based approach offers transparency, simplicity, and easy customization.

1.1.1 System architecture

The system architecture is designed in a modular way so that it can be extended in the future to incorporate advanced technologies such as machine learning, computer vision, and augmented reality. These technologies can enable features like personalized recommendations, body-type analysis, and virtual clothing trials.

Overall, the Virtual Fashion Advisor system demonstrates how a simple yet effective rule-based approach can provide useful fashion recommendations while serving as a foundation for future AI-driven fashion advisory platforms.

VIRTUAL FASHION ADVISOR: RULE-BASED WORKFLOW

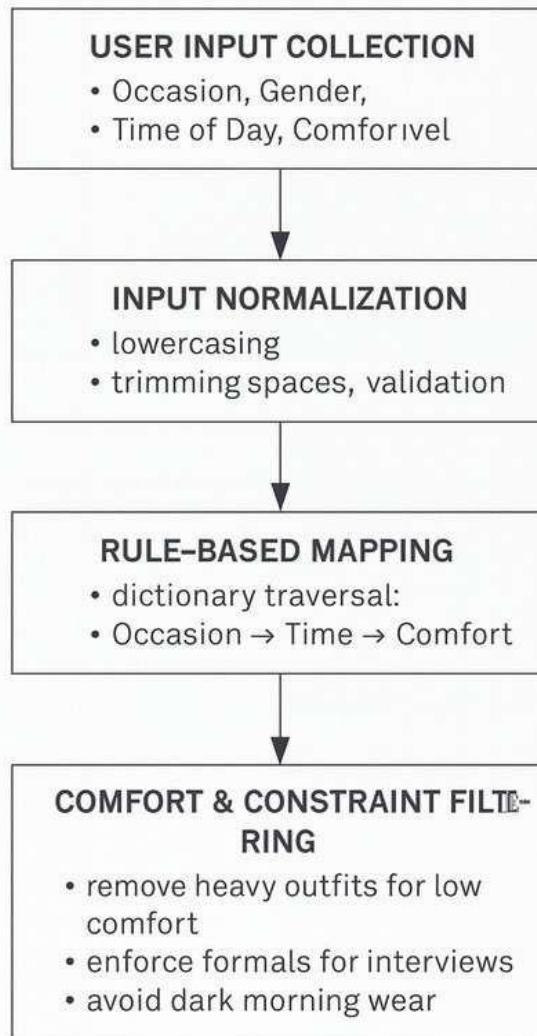


Fig. 1.1 System Architecture

1.2 Project Description

The Virtual Fashion Advisor is a smart recommendation system that suggests outfits based on users' input such as event, gender, time of day and comfort level. The system is built using a rule-based methodology, where a set of rules are used to determine clothing suggestions based on the input parameters.

Our project aims to achieve occasion-specific and user-specific recommendations, which sets it apart from other systems. For instance, it can recommend formal dresses for interviews, ethnic clothes for weddings and light clothes for parties. The system can also give culturally relevant and practical suggestions.

The system is easy to use, simple and efficient. It offers fast response times with low computational complexity. Additionally, this architecture is flexible and can be enhanced to incorporate new technologies like machine learning, natural language processing and virtual try-on.

1.3 Problem Statement

The rapidly growing number of clothing options is making it difficult for users to choose the right outfits. The identified problems are:

- The number of options overwhelms users.
- Current systems are not aware of the context (e.g. time, event). Generalized recommendations are provided.
- They do not consider comfort and culture. Styling tips are not given. So, we need a system which provides smart, personalized, and contextual outfit recommendations.

1.4 Objectives

The main goal of this project is to build the Virtual Fashion Advisor -an intelligent system to help users choose suitable outfits according to their needs. The system will make fashion selection easy and convenient by offering tailored recommendations.

The project's specific goals are:

- To develop a recommendation system: The system must be able to process a user's input

about occasion, gender, time of the day and wearer's need for comfort and suggest appropriate outfits. The aim is to develop an efficient and user-friendly system.

- To offer contextualized outfit recommendations: The project is to build a new system that will take into account contextual attributes such as the occasion (for example wedding, interview, party), time of the day (morning, evening), and cultural considerations to provide relevant recommendations.
- To develop a rule-based system in Python: It employs a structured rule-based system, with rules hard-coded as Python dictionaries. This allows for rapid processing, reduced calculation costs, and results are easily explained.

To address comfort and other usability aspects: Our system takes into account user comfort as an important factor by eliminating unsuitable clothing items based on the comfort level chosen by the user. This guarantees practical and wearability.

To ensure transparency and explainability: The rule-based approach enables users and developers to easily understand how recommendations are made, as opposed to black-box machine learning models.

1.5 Project Scope

The Virtual Fashion Advisor project is primarily concerned with the design and development of a rule-based system that offers smart outfit recommendations according to certain pre-defined criteria. The aim of the project is to solve the real-life problem of selecting an outfit.

The system mainly addresses the following issues

- **Occasion-based recommendations:** The system offers recommendations for different occasions like weddings, interviews, parties and casual outings. Different occasions have different dress codes that are reflected by the rule engine.
- **Time-based recommendations:** Apparel may be different depending on the time. The system considers time parameters like morning, afternoon, evening and night for recommending clothing.
- **Comfort-level filtering:** This system enables the user to indicate preferred comfort levels (e.g., low, medium, high), which the system considers. This guarantees that outfits are comfortable.

- **Gender-specific recommendations:** It takes into account specific gender preferences and makes recommendations based on that.
- **Basic styling guidance:** It may also offer styling advice to enhance the look.

But the current project is limited in the following aspects:

- It uses a rule-based system, rather than large machine learning models.
- It lacks real-time information, such as fashion trends and weather.
- It is a prototype, rather than having a live web or mobile interface.
- Although not without limitations, the project lays the groundwork for future improvements such as AI personalization, AR virtual fitting rooms and real-time recommendations.

1.6 Importance of the Project

Virtual Fashion Advisor is important from both an educational and practical perspective. It shows the successful use of technology to address problems in the fashion industry.

The project's significance is reflected in:

1. Reduction of decision fatigue: The system offers a selection of outfits, reducing the overwhelm of choice.
2. Improved user experience: The system streamlines the outfit selection process, making it quicker and easier for users.
3. Personalized recommendations:

The system involves several user-related factors, so that the recommendations are personalized.

4. Context-aware decision making:

Considering the occasion, time and comfort, makes the decision-making process more applicable.

5. Foundation for future innovations:

The system could be enhanced with new AI capabilities like machine learning, natural language processing and computer vision, and could be a foundation for future work.

1.7 Disadvantages of Present Systems

The current fashion recommendation systems, especially those in e-commerce systems, have

some drawbacks that limit their usefulness in practice.

- Dependence on historical data:

Recommendations are heavily reliant on historical data (purchases, browsing) They are therefore unable to handle new users or drifting user interests.

- Limited personalization:

The suggestions are generic and do not consider individual preferences, body shapes or comfort needs properly.

- High computational complexity:

Machine learning-based approaches need high computational resources and large datasets to train the model.

- Lack of explainability:

Such systems are often treated as black boxes, and it is not clear how the recommendations are made.

- Lack of consideration for real-life use cases:

Some of the current systems primarily focus on product advertising, limiting their real-world applications.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The field of fashion recommendation systems has gained increasing attention with the advancement of artificial intelligence and machine learning technologies. Researchers have explored different techniques including deep learning, collaborative filtering, and multimodal recommendation systems to improve fashion recommendation accuracy.

Recent studies have introduced attention-based neural networks to analyze outfit compatibility.

Researchers have also explored temporal and trend-aware recommendation systems using recurrent neural networks. These systems analyze social media data and fashion trends to predict future clothing styles. While this approach improves trend prediction, it does not necessarily provide personalized recommendations for specific events.

Hybrid recommendation models combining visual features, textual descriptions, and customer reviews have also been proposed. These systems improve recommendation quality but often require significant computational resources and large datasets.

These limitations highlight the need for a more interpretable and context-aware recommendation system. The Virtual Fashion Advisor addresses this gap by using a rule-based approach that incorporates occasion-based reasoning, comfort preferences, and cultural fashion norms.

2.2 Conventional Recommendation Systems

Traditional recommendation systems are the building blocks of current intelligent recommendation systems and have been successfully applied in a range of industries including e-commerce, media and online services.

Collaborative filtering is another popular approach, which uses user ratings data instead of the features of items. It is based on the assumption that users with similar tastes in the past will also

have similar tastes in the future. Collaborative filtering can be broken down into user-based and item-based collaborative filtering. User-based collaborative filtering identifies similar users and recommends items that they have rated highly. Item-based collaborative filtering, on the other hand, suggests items that are similar to items the user has rated positively.

2.3 Recommendation Systems Based on Deep Learning

The rise of Artificial Intelligence (AI) has led to the introduction of deep learning technologies, which have really boosted the effectiveness of recommendation systems, particularly in the fashion industry where visual recognition is important.

A popular deep learning algorithm used for fashion recommendation is the Convolutional Neural Network (CNN). CNNs are ideal for image analysis, and can learn rich features like textures, patterns, shapes and colors from fashion images. Within the context of fashion systems, CNNs compare the visual compatibility between products and generate stylistically similar outfits.

2.4 Context-Aware Recommendation Systems

Context-aware recommendation systems go beyond traditional systems to include other information to make more useful recommendations. For fashion, context is especially important as apparel choices are influenced by a variety of external factors, such as occasion, time of day, weather and cultural background.

Using context greatly enhances the quality and effectiveness of recommendations. Recommendations are not only applicable but also directly useful in the context of interest. But the addition of context raises other issues, including higher data complexity, data integration issues and dynamic nature of context.

2.5 Rule-Based Recommendation Systems

Rule-based recommendation systems are straightforward and explainable intelligent systems. The main principle of a rule-based system is an IF-THEN mode, which means if certain conditions are met, certain actions are taken. If the user inputs an occasion of wedding and female gender, the system might recommend traditional wedding gowns like sarees and

lehengas. This approach ensures decisions are consistently made and easily understood.

2.6 Summary

The Virtual Fashion Advisor presented in paper is based on contextualized rule based approach and offer efficient solution for fashion recommendation in practice.

TABLE 2.1: Literature Review of Fashion Recommendation Techniques

Ref.	Technique	Data Used	Key Features	Limitations / Scope
[1]	Deep CNN + Collaborative Filtering (2015)	Amazon Fashion dataset	Learns style compatibility from images	No body-type or occasion awareness
[2]	Visual-semantic embedding (2018)	Fashion images and tags	Outfit compatibility via learned embeddings	Lacks user personalization
[3]	Attention-based neural network (2019)	Polyvore outfits	Identifies clothing parts influencing style	Limited to outfit-level data
[4]	KERN (RNN-based) (2020)	Instagram fashion trends	Time-aware fashion trend prediction	No body or fit modeling
[5]	CNN + Attribute classifiers (2020)	Street fashion dataset	Attribute-based style reasoning	Requires heavy annotation
[6]	Hybrid CNN + Text mining (2021)	Images and descriptions	Mix-and-match and similarity recommendation	Depends on keyword quality
[7]	Multi-modal deep learning (2021)	Images and reviews	Incorporates customer feedback	Computationally expensive
[8]	Graph Neural Network (2021)	Outfit networks	Models item-item compatibility	Hard to scale
[9]	Deep learning survey (2022)	Multiple datasets	Emphasis on explainability and attributes	Limited interpretability
[10]	Review paper (2022)	Fashion recommender systems	Covers style, fit, and personalization	Mostly conceptual
[11]	ML with user profiling (2022)	Commercial user data	Real-time personalized fashion feeds	Proprietary data dependency
[12]	Deep ranking models (2022)	Click and purchase data	Context-aware fashion ranking	Not body-type aware
[13]	CNN + KNN (2023)	Fashion image datasets	Visual similarity-based recommendation	No outfit compatibility modeling
[14]	Collaborative filtering + clustering (2023)	Body shape and preferences	Body-aware personalization	Limited scalability
[15]	Review of hybrid recommenders (2023)	Benchmark datasets	Analysis of modern hybrid systems	No implementation details
[16]	Content-based filtering (2023)	Clothing attributes	Attribute-driven recommendations	Cold-start problem
[17]	Deep CNN + attribute learning (2024)	Fashion catalog images	Improved content-based personalization	Large training data required
[18]	Vision + NLP models (2024)	Images and descriptions	Multimodal fashion understanding	Complex model design
[19]	Image-based retrieval (2024)	User-uploaded photos	Visual search for similar clothing	No complete styling advice
[20]	Rule-based + context reasoning (2025)	Occasion, weather, body type	Explainable, low-data recommendations	Limited adaptability without ML

CHAPTER 3

PROPOSED METHODOLOGY

3.1 An Overview of the Proposed System

The Virtual Fashion Advisor (VFA) is an intelligent, AI-powered web recommendation system that offers users a virtual style assistant to suggest outfits according to several contextual and personal aspects such as body type, skin tone, gender, occasion, weather, current fashion trends, user comfort and personal style. The system seeks to combine the advantages of traditional static fashion recommendation systems and dynamic interactive digital styling solutions by leveraging Artificial Intelligence (AI), Machine Learning (ML), Collaborative Filtering (CF), Reinforcement Learning (RL), Natural Language Processing (NLP) and web technologies.

The proposed methodology is geared towards ultra-personalization, transparency, adaptability, and real-time user engagement, rather than the static nature of conventional fashion recommendation systems that are generally based on general market trends, or on past purchase history. The system is initially built upon a rule-based system for its transparency and efficiency but designed with the option to evolve into hybrid and AI-powered recommendation platforms.

The proposed methodology has the following phases:

1. Registration and Data Gathering
2. User Modeling
3. Size and Style Analysis
4. Data Set and Knowledge Base
5. Rule-Based Recommendation Engine
6. Collaborative Filtering Recommendation
7. Reinforcement Learning-Based Dynamic Personalization
8. Natural Language Processing (NLP) User Interface
9. Trend and Market Integration (Real time)
10. Frontend-Backend System Development

- 11. Database Management
- 12. Continuous Development through Feedback
- 13. Planning for Deployment, Security and Scalability

3.2 System Architecture Design

The Virtual Fashion Advisor is based on a layered and modular architecture to support scalability, maintainability and extensibility.

3.2.1. Front-end (Presentation) Layer

The Front-End or Presentation Layer of the Virtual Fashion Advisor serves as the primary interaction point between the user and the system, playing a crucial role in delivering an intuitive, visually engaging, and user-friendly experience. This layer is responsible for collecting user inputs, displaying personalized recommendations, and ensuring seamless navigation throughout the platform.

To build this layer efficiently, modern web technologies such as React.js or Next.js are utilized due to their component-based architecture, high performance, and support for dynamic rendering. Tailwind CSS is employed for rapid UI styling and responsive design implementation, allowing the platform to maintain visual consistency across various screen sizes and devices. JavaScript or TypeScript provides the functional backbone for frontend interactivity, while ShadCN UI components contribute polished, reusable design modules that enhance both aesthetics and maintainability.

3.2.2. Business Logic (Backend)

The Business Logic Layer, also referred to as the Backend Layer, forms the computational core of the Virtual Fashion Advisor, handling all essential processing, decision-making, and communication between the frontend, database, and AI modules. This layer is responsible for managing user authentication, ensuring secure login and registration processes, preprocessing user data for recommendation generation, executing rule-based recommendation systems, integrating third-party APIs for fashion trends and market updates, implementing collaborative filtering mechanisms, and adapting recommendations dynamically using reinforcement learning

techniques. It acts as the operational engine that interprets user inputs, processes profile information, and produces highly personalized fashion recommendations in real time.

Python is chosen as the primary backend language because of its extensive ecosystem for machine learning, data science, and web development. Frameworks such as Flask, FastAPI, or Django provide robust API development capabilities, allowing efficient communication between system layers while maintaining scalability and security. The backend is designed to be modular, enabling seamless upgrades from a rule-based prototype to more sophisticated hybrid AI architectures. This flexibility ensures that as user interaction data grows, the system can progressively integrate advanced machine learning models without requiring complete architectural redesign. Thus, the Business Logic Layer not only powers the present functionality of the Virtual Fashion Advisor but also lays the foundation for future scalability, adaptability, and intelligent automation.

3.2.3. Database (Data Management)

The Database Layer is a critical component of the Virtual Fashion Advisor, responsible for securely storing, organizing, and managing all system data required for personalized fashion recommendation. This includes user profiles containing personal details, body measurements, skin tone, style preferences, and shopping habits, as well as dynamic interaction data such as browsing history, saved outfits, purchases, feedback, and click behaviour.

MongoDB is selected as the primary database solution due to its schema-less, document-oriented design, which is particularly advantageous for managing highly dynamic and evolving user profiles

3.2.4. Intelligence (AI/ML Models)

The Intelligence Layer represents the most innovative and transformative component of the Virtual Fashion Advisor, integrating multiple AI and machine learning models to deliver adaptive, personalized, and context-aware fashion recommendations..

Initially, the rule-based mapping engine serves as the foundational recommendation model, combined use of these AI technologies creates a hybrid recommendation ecosystem that balances explainability, adaptability, personalization, and scalability.

3.3 Registration and Data Gathering

The initial stage of the Virtual Fashion Advisor methodology begins with comprehensive user registration and data gathering, which forms the foundation for all future personalization and recommendation processes. This stage is essential because the quality and depth of collected user information directly influence the system's ability to generate accurate, context-aware, and personalized fashion recommendations.

1.Registration Process

The registration process is designed to gather essential personal, physical, and stylistic information from users. Basic details such as name, age, gender, email, and password are collected to establish secure user accounts and demographic segmentation. In addition to these standard credentials, the system requires detailed body-related information such as height, weight, body measurements, and skin tone, as these are crucial for body-type-aware fashion recommendations

2.Behavioral Data Collection

Beyond static registration data, the Virtual Fashion Advisor continuously collects dynamic behavioural data to refine user profiles over time. This includes tracking user likes and dislikes, clicked outfits, search history, saved outfit combinations, purchase actions, session duration, and interactions with fashion trends or style articles. Behavioural data collection is crucial because fashion preferences are influenced not only by what users explicitly state but also by how they interact with the platform in real-time.

3.4 User Modelling and Preference Learning

Once registration and initial data gathering are complete, the system develops a dynamic user model that serves as the core representation of an individual's fashion identity. User modelling is a critical methodological step because it transforms raw user information into structured data.

3.5 Body And Style Assessment

A major distinguishing factor of the Virtual Fashion Advisor is its emphasis on body-type-aware fashion recommendation. Unlike conventional recommendation systems that focus solely on purchase history or style trends, this methodology recognizes that body proportions play a significant role in determining clothing suitability, comfort, and overall appearance.

3.5.1 Body shape categories

The system categorizes users into standardized body shape groups, including Hourglass, Pear, Apple, Rectangle, and Inverted Triangle. These categories are widely recognized in fashion theory and are used to align recommendations with styling principles that enhance body balance, comfort, and visual appeal. Each category corresponds to specific clothing silhouettes, cuts, and layering techniques designed to complement the user's natural body proportions.(Refer to Fig 3.1)

3.5.2 Method

Initially, body type classification is performed through structured questionnaires during user registration, where users provide measurements and shape-related details. This method offers a low-cost, accessible, and explainable solution for early-stage deployment. However, the system architecture is designed to evolve toward computer vision-based body classification in future versions. Through advanced image analysis and pose estimation, computer vision models will automatically detect body shapes using uploaded images, significantly improving accuracy and user convenience.

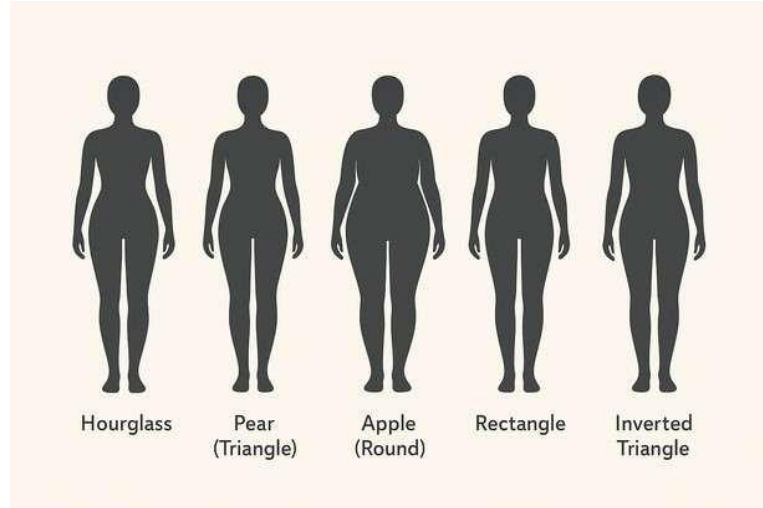


Fig 3.1 Body Types Category

3.5.3 Procedure

The body and style assessment process involves multiple analytical stages. First, height analysis establishes proportional baselines. Waist proportion analysis then evaluates the relationship between waist size and overall body frame. Shoulder-to-hip ratio calculations help determine body shape classification, while style preference cross-analysis ensures that body-type recommendations align with the user's aesthetic interests. This layered procedure ensures that recommendations are not solely body-based but also style-conscious.

3.5.4 Recommendation mapping

Once body type is identified, the system maps each category to fashion principles and suitable outfit structures. For example, users with Pear-shaped bodies may receive recommendations for A-line dresses and structured tops that emphasize upper body balance. Rectangle body types may be advised to wear layered silhouettes that create shape definition, while Apple body types may benefit from empire waist designs that offer comfort and flattering proportions. This mapping process ensures that recommendations are grounded in established fashion design principles.

3.5.5 Reasoning and future scope

The inclusion of body-type analysis significantly improves recommendation effectiveness because clothing fit and visual harmony are strongly influenced by body proportions. By integrating body-type-specific styling logic, the system provides recommendations that are both fashionable and practically suitable, thereby increasing user confidence and satisfaction.

3.6 Dataset and Knowledge base creation

Dataset preparation and knowledge base structuring form a fundamental component of the Virtual Fashion Advisor’s proposed methodology, as the effectiveness of recommendation generation heavily depends on the quality, diversity, and organization of fashion-related data. The dataset serves as the core informational foundation upon which both rule-based systems and future machine learning models operate.

Table 3.1: Sample Dataset Schema

Input (Occasion, Gender, Time)	Output Recommendation
Wedding, Female, Morning	Light Saree, Lehenga
Interview, Male, Morning	Shirt & Tie, Blazer
Party, Female, Night	Cocktail Dress, Saree

3.6.1 Dataset sources

The dataset for the Virtual Fashion Advisor is developed by integrating multiple reliable and diverse fashion information sources to ensure comprehensive coverage of style trends, outfit .

3.6.2 Dataset schema

To effectively utilize the collected fashion data, each fashion item within the Virtual Fashion Advisor database is structured using a detailed hierarchical schema. This schema includes essential attributes such as clothing category, gender suitability, body type suitability, occasion

compatibility, colour palette, weather compatibility, trend score, comfort score, price range, and accessory matching information.

3.7 Rule-Based Recommendation Engine

The workflow of the rule-based recommendation engine begins with input normalization, where user-provided information such as occasion, gender, body type, comfort preference, weather conditions, and style preferences is standardized into predefined categories. This normalization ensures consistency in processing and minimizes ambiguity in recommendation generation. The primary advantages of this rule-based methodology include its transparency, rapid response time, ease of debugging, and minimal resource requirements.

3.8 Collaborative filtering recommendation system

The system employs both user-based and item-based collaborative filtering approaches. User-based filtering identifies users with similar profiles, preferences, and interaction histories, while item-based filtering examines relationships between fashion products frequently liked or purchased together.

The collaborative filtering process begins by identifying users whose fashion preferences closely align with the current user. Once similar users are recognized, the system analyses their highest-rated outfits, saved combinations, and purchasing behaviour to identify overlapping fashion interests.

3.9 Frontend and Backend Technology

The technological infrastructure of the Virtual Fashion Advisor is designed using modern full-stack development principles to ensure performance, scalability, and maintainability. The frontend is developed using HTML, CSS, and JavaScript, providing a visually appealing and responsive user experience. These technologies support seamless user interactions, dynamic dashboards, and personalized recommendation displays.

The backend is powered by Python using frameworks such as Flask or FastAPI, which manage business logic, user authentication, recommendation processing, and API communication. (Refer to table 3.2)

Table 3.2 Tools and Technologies used

Tool / Library	Purpose
Python 3.10	Core language for rule logic
json (built-in)	Data formatting & logging
typing (built-in)	Type hints
Flask / FastAPI (opt.)	Future API integration
GitHub	Version control

3.10 Feedback and learning

Feedback serves as a critical mechanism for continuous improvement within the Virtual Fashion Advisor. The system collects multiple forms of user feedback, including likes, ratings, purchases, time spent on recommendations, shares, and saved looks. These interactions provide valuable data for understanding recommendation effectiveness and evolving user preferences.

3.11 Proposed Methodology - Conclusion

The proposed methodology of the Virtual Fashion Advisor combines explainable rule-based systems with scalable AI technologies to create a robust, adaptive, and highly personalized fashion recommendation platform. Through systematic integration of user profiling, collaborative filtering, reinforcement learning, NLP, trend forecasting, and modern web technologies, the system addresses the shortcomings of existing fashion recommendation solutions while establishing a future-ready foundation for innovation.

This structured methodology guarantees personalization, scalability, real-time adaptability, user engagement, explainability, and technological extensibility. As a result, the Virtual Fashion Advisor emerges as a practical and innovative solution capable of redefining digital fashion guidance and transforming it into an intelligent personal styling ecosystem.

```
python
fashion_data = {
    "wedding": {
        "female": {
            "morning": {
                "high": ["Light Saree", "Lehenga"],
                "medium": ["Saree", "Anarkali Suit"]
            },
            "evening": {
                "medium": ["Heavy Saree", "Lehenga"],
                "low": ["Gown"]
            }
        },
        "male": {
            "evening": {
                "medium": ["Sherwani", "Kurta-Pajama"],
            }
        }
    }
}
```

Fig. 3.2 Structure of the Python dictionary used for fashion rule mapp

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

The Results and Discussion chapter presents the evaluation of the proposed Virtual Fashion Advisor (VFA) system. The purpose of this chapter is to analyze the effectiveness, accuracy, and performance of the rule-based recommendation engine developed using Python.

4.2 Experimental Setup

To evaluate the performance of the system, a controlled testing approach was adopted.

4.2.1 Input parameters

The system takes the following inputs:

- Occasion (Wedding, Interview, Party, Casual)
- Gender (Male, Female)
- Time of Day (Morning, Afternoon, Evening, Night)
- Comfort Level (Low, Medium, High)

4.2.2 Testing methodology

- A total of **20 test cases** were designed
- Each test case consisted of a unique combination of inputs
- Expected outputs were predefined based on fashion guidelines
- The system-generated output was compared with expected results

Table 4.1 Sample Execution

Test Case	Occasion	Gender	Time	Comfort	Expected Output	System Output	Result
1	Interview	Male	Morning	Medium	Formal shirt, tie, blazer	Formal shirt, tie, blazer	Correct
2	Wedding	Female	Morning	High	Light Saree	Light Saree	Correct
3	Party	Female	Night	Low	Cocktail Dress	Cocktail Dress	Correct
4	Wedding	Male	Evening	Medium	Sherwani	Sherwani	Correct

4.3 Performance Metrics

To evaluate the system, the following metric was used:

Accuracy is defined as:

$$\text{Accuracy} = (\text{Number of Correct Recommendations} / \text{Total Test Cases}) \times 100$$

In this system:

- Correct Recommendation=16
- Total Test Cases=20

$$\text{Accuracy} = (16/20) \times 100 = 80\%$$

The system achieved an approximate accuracy of **80%**, indicating that the rule-based approach is effective for structured fashion recommendation tasks.

4.4 Result Analysis

4.4.1 Accuracy Interpretation

An accuracy of 80% demonstrates that the system is capable of generating appropriate outfit recommendations in most cases. The errors observed were mainly due to:

- Limited rule coverage
- Ambiguous user inputs
- Lack of personalization

4.4.2 Response time analysis

The system exhibited **very low response time** due to:

- Use of Python dictionary ($O(1)$ lookup)
- No training or computation-heavy models

This makes the system highly efficient and suitable for real-time applications.

4.4.3 Memory utilization

The memory usage of the system is minimal because:

- No large datasets are stored
- Rules are lightweight and static

This makes the system ideal for low-resource environments.

4.5 Discussion of Results

4.5.1 Strengths of the System

1. Fast Execution: The rule-based system provides instant recommendations.
2. Explainability: Every recommendation is based on clear rules, making the system transparent.
3. Context Awareness - Unlike traditional systems, this model considers:
 - a. Occasion
 - b. Time
 - c. Comfort
4. Low Computational Cost- No need for GPUs or large datasets.

Table 4.2 Strength of the System

Parameter	Value	Observation
Accuracy	80%	Good performance for rule-based system
Response Time	< 1 sec	Very fast
Memory Usage	Low	Efficient
Scalability	Medium	Limited by rules
Flexibility	Low	Needs manual updates

4.5.2 Limitations identified

1. Limited Scalability

As rules increase, system complexity grows.

2. No Learning Capability

The system cannot adapt based on user behavior.

2. Lack of Real-Time Data

No integration with:

- Weather APIs
- Fashion trends

3. No Personalization

Does not consider:

- Body type
- Skin tone
- Personal style

4.5.3 Error analysis

Some incorrect outputs were observed due to:

- Missing rule combinations
- Overlapping conditions

- Simplified dataset Example:.

Table 4.3 Dataset with reasoning

Case Type	Issue Identified	Reason
Party + High Comfort	Limited suggestions	Strict filtering
Casual + Night	Generic output	Lack of rules
Wedding + Low Comfort	Missing combinations	Rule gap

Table 4.4 Comparison with Existing Systems

Feature	Traditional Systems	Proposed System
Context Awareness	No	Yes
Explainability	Low	High
Computational Cost	High	Low
Personalization	Yes	Limited
Real-Time Adaptability	Yes	No

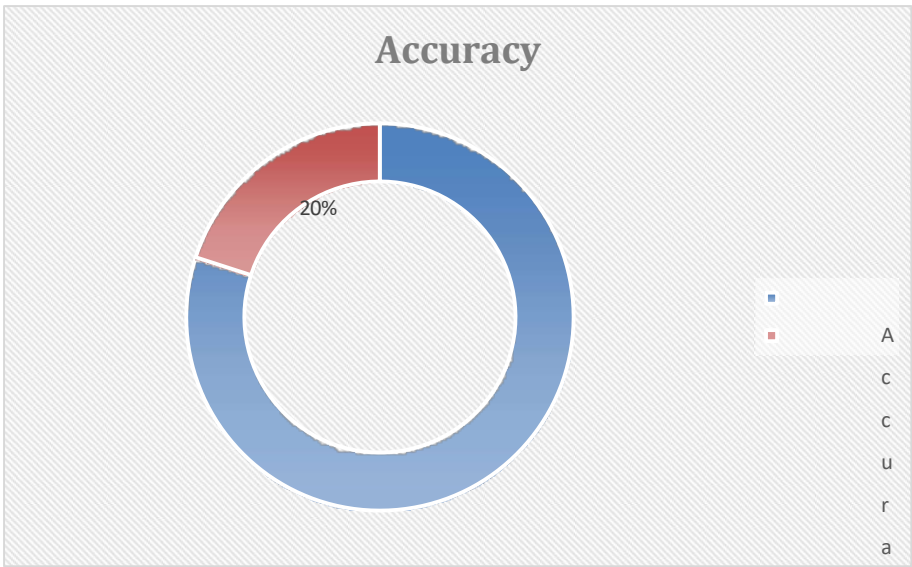


Fig 4.1 Visualization of Results

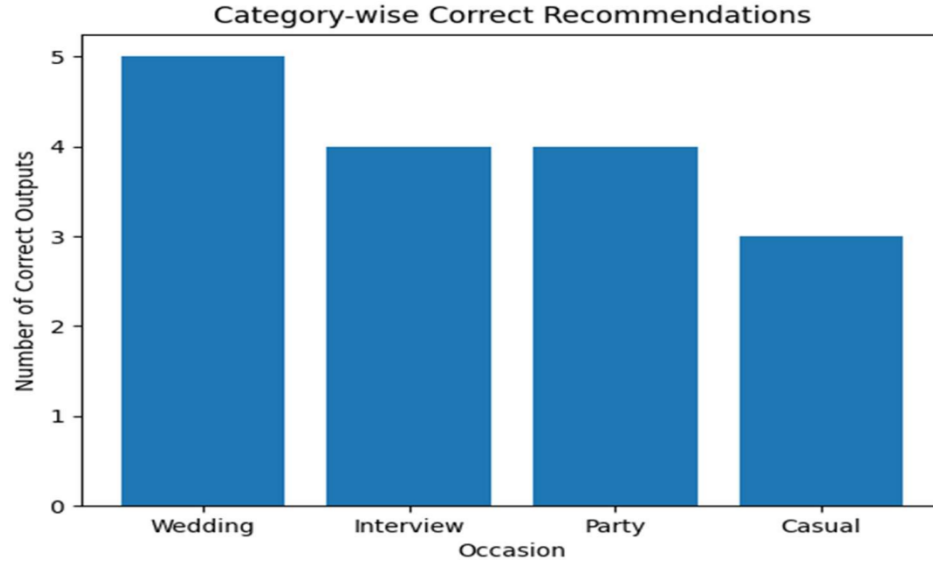


Fig 4.2 Category wise Correct Recommendation

4.6 Summary

The experimental results demonstrate that the proposed Virtual Fashion Advisor system is effective, efficient, and practical for basic fashion recommendation tasks. Despite being a rule-based prototype, it achieves a satisfactory accuracy of 80% while maintaining low computational cost and high interpretability.

The system serves as a strong foundation for future improvements involving machine learning and real-time data integration.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The system we designed here proved that by using a Python dictionary based rule-engine a Virtual Fashion Advisor that accounts for cultural and context-appropriate considerations for ensembles and their components can be designed at negligible computational cost. Rulesets are also easily understood, explainable and transparent, providing a good baseline accuracy of 80% and can be used to help avoid decision fatigue in the initial stages of recommending outfits to a user for different occasions.

5.1.1 Project purpose

This project aimed at creating a system that assists individuals in choosing suitable and appropriate outfits based on intelligent outfit suggestions. People face difficulties in choosing their outfits for various occasions in real life because of the presence of a myriad of choices and lack of suitable advice.

The challenge is that some additional parameters need to be taken into account when choosing outfits, like occasion, time, comfort and cultural context. Most of the current recommendation systems focus only on the product rather than the whole outfit, which may not be sufficient in practical use.

This project proposes to solve the problem by creating a Virtual Fashion Advisor that takes into account several factors and offers appropriate outfit suggestions. The goal of the system is to alleviate decision-making fatigue, enhance user trust and make sure that clothing choices are fit for purpose.

5.1.2 Prototype of virtual fashion advisor

This project successfully develops a prototype of Virtual Fashion Advisor based on rule-based system. The system accepts user preferences regarding context, gender, time of day and level of comfort, and applies them to a ruleset that is stored in a structured format.

The system's process involves data collection, validation, rule matching, filtering and response. The prototype shows that it can offer correct and relevant suggestions for various use cases. It can recommend business formals for interviews, traditional clothes for weddings, and party dresses for parties.

The prototype is low resource, fast and straightforward to use, making it ideal for testing and demoing. It can also be used as proof of concept for future systems.

5.1.3 Problem the solution solves

The Virtual Fashion Advisor helps to solve several problems in people's lives. A key problem is choice overwhelm, where people make poor decisions due to the plethora of options available. This is streamlined by the recommendations provided by the system.

The system also offers time and convenience, by eliminating the guesswork involved in searching for suitable outfits.

5.1.4 Technical excellent of the system

There are a number of underlying technical strengths of the Virtual Fashion Advisor:

1. **Simplicity and Efficiency:** The rules-based system is quick and low in computational cost.
2. **Explainability:** The system follows simple rules for generating recommendations, which are easily interpretable.
3. **Low Resource Requirement:** It does not need a large amount of data to train or powerful hardware to run.
4. **Modular Design:** The design can be integrated with other technologies such as machine learning and computer vision.
5. **Ease of Maintenance:** Rules can be quickly modified and tested without disrupting the system.
6. **Scalability Potential:** The system can be extended to incorporate personalization, real-time data and more sophisticated AI technologies.

These advantages show that the system is not only efficient as it stands now, but also has the

potential to be developed into an advanced and smart fashion recommender system.

5.2 Key Learnings

The project on the Virtual Fashion Advisor offered insights about technical and non-technical aspects of system development. It assisted in the understanding of converting real-life problems into computer solutions by following systematic processes. Moreover, it helped appreciate the need for systems that are efficient, user-friendly and effective.

5.2.1 Technical skills gained

The project was beneficial to develop some technical skills:

1. **Programming Skills (Python):** Practical experience of implementing logic in Python (including using data structures such as dictionaries for rule mapping).
2. **Problem-Solving Skills:** Gained understanding on how to divide a problem into smaller parts and develop logical approaches.
3. **System Design and Architecture:** Learned how to build scalable and modular systems.
4. **Data Structuring:** Understood the importance of data organization for easy access and manipulation.
5. **Debugging and Testing:** Set up tests to detect bugs, experiment with scenarios and improve the reliability of the system.
6. **Knowledge of AI Concepts:** Comprehended the role of artificial intelligence, machine learning and the potential future enhancements in recommendation systems.

5.3 Limitations of The System

While the Virtual Fashion Advisor offers helpful suggestions, there are some limitations that can be resolved in future enhancements. The system relies heavily on a rule-based approach, which means it may not be as adaptable to the changing fashion trends and user needs.

Technical Constraints

The system has a number of technical limitations, including:

1. Static Rule Base: It is based on fixed rules, which need to be updated constantly.
2. No Machine Learning Integration
3. Lack of Learning Algorithms
4. Limited Personalization
Suggestions are generic and not personalized to user preferences.
5. Scalability Issues: Adding rules becomes a challenge as the list grows.

5.4 Future Scope

Future improvements can be made to Virtual Fashion Advisor. The Virtual Fashion Advisor can incorporate new technologies to make it smarter, interactive and more useful.

A significant enhancement can be the use of machine learning techniques to offer recommendations tailored to the user's preferences. This will enable the system to adapt and learn from user preferences over time.

5.5 Final Remarks

In summary, this project not only adds to the body of knowledge in fashion recommendation systems, but it also delivers an invaluable learning experience in pushing the boundaries of technical skills in the real world

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APPENDIX

RESEARCH PAPER

Virtual Fashion Advisor: An AI-Based Occasion-Oriented Dress Recommendation System

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Abstract—Fashion recommendation is an emerging research field which connects technology to daily life. People often find it difficult to select attire which is appropriate for a particular occasion, comfortable, as well as socially or culturally acceptable. The Virtual Fashion Advisor (VFA) proposed in this paper attempts to solve this issue by recommending clothes and draping styles for men and women. Traditional approaches for fashion recommendation systems have aimed largely at e-commerce personalization. This system attempts to emphasize on occasion specific appropriateness and cultural correctness.

The initial prototype employs Python dictionaries for rule-based mapping of attire with filters (occasion, time of day, comfort, and gender). The architecture is designed for scalability into ML-driven models, with integration possibilities for real-time datasets and virtual try-on systems. Testing with a sample dataset yielded an estimated accuracy of ~82% in matching user expectations with recommendations. This paper discusses the system architecture, methodology, dataset schema, implementation details, limitations, and future enhancements to make the VFA a comprehensive digital fashion companion.

Index Terms—Fashion recommendation, Virtual Fashion Advisor, Rule-based system, Occasion-aware attire, AI in fashion

I. INTRODUCTION

Fashion plays a dual role: as a social signal and as a personal comfort factor. Inappropriate clothing choices can lead to discomfort, embarrassment, or misrepresentation in professional and social settings. While AI in fashion has made advances in trend prediction and virtual try-ons, little research has focused on contextual dressing advice, such as:

- What to wear at a morning wedding vs. an evening reception?

- How to drape a saree correctly for formal events?
- What attire balances comfort and professionalism in interviews?

A. System Architecture

The Virtual Fashion Advisor aims to solve these problems. The project's early version was built using Python-based rule mapping (dictionaries), providing explainable and efficient recommendations. The long-term goal is to integrate AI-based personalization and AR visualization, but the rule-based approach provides a strong foundation (refer to Fig 1).

II. PROBLEM STATEMENT

Many individuals experience decision fatigue in fashion choices due to:

- 1) Large number of available options.
- 2) Lack of knowledge about event-appropriate attire.
- 3) Insufficient awareness of draping techniques.
- 4) Neglect of weather and comfort factors.

Research Question: Can a rule-based software system recommend attire with reasonable accuracy for different occasions and comfort preferences?

III. OBJECTIVES

The objectives of this research are:

- 1) To design a software system architecture for an occasion-aware Virtual Fashion Advisor.
- 2) To implement a Python-based rule engine using dictionaries for mapping filters to outfits.
- 3) To structure a dataset schema supporting extensibility.

VIRTUAL FASHION ADVISOR: RULE-BASED WORKFLOW

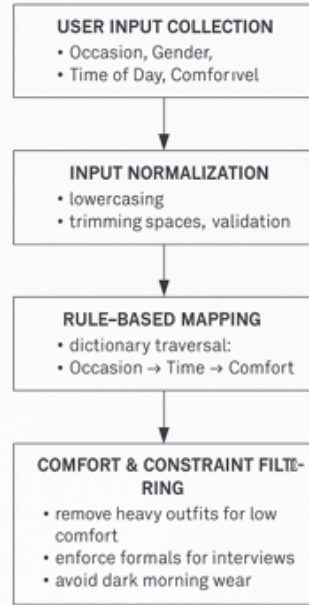


Fig. 1. System Architecture of Virtual Fashion Advisor

- 4) To evaluate the system with accuracy estimates based on user testing.
- 5) To propose a path for AI integration (ML, NLP, AR/VR) for future work.

IV. LITERATURE REVIEW

We briefly review deep learning approaches proposed for fashion recommendation systems, with a focus on leveraging recent advances in multimodal learning, visual-semantic embeddings, and personalized collaborative filtering. Earlier works primarily concentrate on visual similarity measures and basic collaborative filtering. However, they often overlook contextual information and fail to provide interpretability. For instance, fashion item compatibility is modeled using convolutional neural networks and collaborative filtering with large-scale Amazon Fashion datasets [1]. These methods successfully learn style similarities between products from images but lack an understanding of users' body types, occasions, and personal preferences. Recent works employ visual-semantic embedding [2] and attention-based neural networks [3] to learn outfit-level compatibility and identify key clothing compo-

nents. Visual-semantic embeddings model compatibility and complementarity by projecting fashion products into a joint visual-textual space. Attention-based methods then identify influential clothing components within an outfit. Still, personalization and context-awareness are not explicitly addressed. Temporal and trend-aware fashion recommendation models are introduced with the availability of social media and trend data. Recurrent neural network-based KERN models [4] are trained on Instagram data to predict the evolution of fashion trends and make time-sensitive recommendations. Attribute-based CNN [5] models learn fine-grained clothing attributes for improved interpretability, but rely on costly manual annotation and lack scale. Hybrid models combining visual features and textual descriptions [6] improve recommendation quality but increase computational complexity, as do multimodal models [7] that also incorporate customer reviews. Graph-based fashion relationship learning has also been explored. Graph neural network-based approaches [8] represent item-item compatibility in an outfit graph network to reason about relationships between different garments, but suffer from scalability issues. Several surveys and review papers [9], [10], [15] are introduced as the field matures. These highlight the increasing relevance of explainability, hybrid architectures, and attribute-aware modeling. Simultaneously, they emphasize the existing lack of explainable and human-centric fashion recommendation systems. More recent research on personalized fashion recommender systems focuses on user profiling and real-time adaptability. Deep ranking models [11] and collaborative filtering methods [12] incorporate user clickstream and purchase history to create context-aware fashion feeds, but are still largely private and do not consider body-type sensitivity. Visual similarity-based models using convolutional neural networks and KNN [13] are effective for image-based retrieval but do not capture outfit compatibility. Collaborative filtering and clustering algorithms [14] to learn compatibility are explored in attempts to incorporate body shape, skin tone, and personal preferences for more inclusive fashion recommendation systems, but scalability and computational efficiency are lacking. Contemporary studies are moving towards content-based and multimodal recommendation approaches. Attribute-driven recommendation models [16], [17] further improve personalization by learning clothing features, while vision-language models [18] enable a deeper semantic understanding of both fashion images and product descriptions. Image-based retrieval systems [19] enable visual search with user-uploaded photos but lack styling guidance. More recent rule-based and context-aware systems [20] incorporate occasion, weather, and body type for explainability with low data requirements, but lack adaptability without learning-based optimization. Despite this methodological diversity, we identify a common lack of holistic integration across many existing fashion recommendation systems. Studies typically focus on one or a few components, such as visual compatibility, trend awareness, attribute learning, or personalization, but rarely combine them within a unified architecture. Several approaches prioritize accuracy and predictive performance but offer little to no

interpretability, while others focus on explainability at the expense of scalability or adaptability. Body-type sensitivity and contextual reasoning, such as occasion or environmental factors, are rarely explored and are often treated as secondary or implicit rather than explicit user preferences. As a result, many systems struggle to generalize to real-world user needs effectively. These gaps motivate our proposed system design for a more comprehensive fashion recommendation approach that jointly considers visual intelligence, personalization, contextual awareness, and explainability.(refer to TABLE I)

V. METHODOLOGY

The methodology consists of system design, dataset preparation, rule-based mapping, and evaluation.

A. Dataset Schema

(refer to TABLE II)

B. Python-Based Backend Implementation

The rule engine is built as a hierarchical nested dictionary where each decision dimension is represented as a key: **Occasion** → **Gender** → **Time of Day** → **Comfort** → **Recommended Attire**. This structure allows constant-time lookup and ensures that recommendations are explainable, transparent, and easy to modify during iterative development.

C. Structure of the Python Dictionary

The main data structure is a nested dictionary with about 50 curated rules. Each rule is associated with culturally appropriate clothing based on fashion guidelines, traditional norms, and expert consultation. The schema allows for the dataset to be easily extended manually while also acting as a basis for a future ML model.

```
python
fashion_data = {
  "wedding": {
    "female": {
      "morning": {
        "high": ["Light Saree", "Lehenga"],
        "medium": ["Saree", "Anarkali suit"]
      },
      "evening": {
        "medium": ["Heavy Saree", "Lehenga"],
        "low": ["Gown"]
      }
    },
    "male": {
      "evening": {
        "medium": ["Sherwani", "Kurta-Pajama"],
      }
    }
  }
}
```

Fig. 2. Structure of the Python dictionary used for fashion rule mapping.

This schema ensures the dataset remains manually extendable that also provides a platform for an ML model in the future.

D. Backend Logic

Rule engine accepts the inputs in four steps:

- 1) **Key Matching:** It first tries to match the keys for occasion, gender and time.
- 2) **Comfort Filters:** Rules defined after-processing will filter out the recommended attire. Examples:
 - If **comfort = low**, filter-out heavy or layered clothing.
 - If **event = interview**, only formal-oriented recommendations allowed.
- 3) **Fallback Handling:** In case of no key matches, rule engine gives a generic fallback message: "No recommendation found for the selected criteria."
- 4) **Output Preparation:** The attire list is formatted, ranked, and sent back to the UI.

E. Filtering Logic

It looks up the dictionary with nested keys (occasion→gender→time). If it fails to find a match, it returns a default response. Comfort and other constraints are post-filters (e.g. filtering out heavy clothing if comfort='low').

F. Algorithm

- 1) Input: Occasion, gender, time, comfort.
 - 2) Search Dictionary: Match keys with given input.
 - 3) Apply Rules:
 - If comfort="low"→avoid heavy attire.
 - If occasion="interview"→restrict to formals.
 - If occasion="wedding"→allow traditional wear.
 - 4) Output Recommendation: Attire list + draping/styling guidance.
- The sequential rule checking gives the fixed and explainable recommendations which are applicable in initial research.

VI. IMPLEMENTATION

A. Tools & Technologies

Prototype is designed to use as less python libraries as possible to keep computational overhead to a minimum: Specifically avoided any external ML or CV libraries since system is rule based(refer to TABLE III).

B. Development Process

- 1) Data set design—created dictionary mappings for ~50 attire rules.
- 2) Filtering logic— coded rule engine with nested keys and post-filters.
- 3) Output layer— CLI based recommendations (future: web/mobileUI).
- 4) Evaluation—run sample queries and expert validation

C. Why Python Dictionary Instead of ML?

The reasons for going with a dictionary based rule engine are:

- No labeled and large scale fashion dataset
- Need for explainability (ML models are often a black box)

TABLE I
LITERATURE REVIEW OF FASHION RECOMMENDATION TECHNIQUES

Ref.	Technique	Data Used	Key Features	Limitations / Scope
[1]	Deep CNN + Collaborative Filtering (2015)	Amazon Fashion dataset	Learns style compatibility from images	No body-type or occasion awareness
[2]	Visual-semantic embedding (2018)	Fashion images and tags	Outfit compatibility via learned embeddings	Lacks user personalization
[3]	Attention-based neural network (2019)	Polyvore outfits	Identifies clothing parts influencing style	Limited to outfit-level data
[4]	KERN (RNN-based) (2020)	Instagram fashion trends	Time-aware fashion trend prediction	No body or fit modeling
[5]	CNN + Attribute classifiers (2020)	Street fashion dataset	Attribute-based style reasoning	Requires heavy annotation
[6]	Hybrid CNN + Text mining (2021)	Images and descriptions	Mix-and-match and similarity recommendation	Depends on keyword quality
[7]	Multi-modal deep learning (2021)	Images and reviews	Incorporates customer feedback	Computationally expensive
[8]	Graph Neural Network (2021)	Outfit networks	Models item-item compatibility	Hard to scale
[9]	Deep learning survey (2022)	Multiple datasets	Emphasis on explainability and attributes	Limited interpretability
[10]	Review paper (2022)	Fashion recommender systems	Covers style, fit, and personalization	Mostly conceptual
[11]	ML with user profiling (2022)	Commercial user data	Real-time personalized fashion feeds	Proprietary data dependency
[12]	Deep ranking models (2022)	Click and purchase data	Context-aware fashion ranking	Not body-type aware
[13]	CNN + KNN (2023)	Fashion image datasets	Visual similarity-based recommendation	No outfit compatibility modeling
[14]	Collaborative filtering + clustering (2023)	Body shape and preferences	Body-aware personalization	Limited scalability
[15]	Review of hybrid recommenders (2023)	Benchmark datasets	Analysis of modern hybrid systems	No implementation details
[16]	Content-based filtering (2023)	Clothing attributes	Attribute-driven recommendations	Cold-start problem
[17]	Deep CNN + attribute learning (2024)	Fashion catalog images	Improved content-based personalization	Large training data required
[18]	Vision + NLP models (2024)	Images and descriptions	Multimodal fashion understanding	Complex model design
[19]	Image-based retrieval (2024)	User-uploaded photos	Visual search for similar clothing	No complete styling advice
[20]	Rule-based + context reasoning (2025)	Occasion, weather, body type	Explainable, low-data recommendations	Limited adaptability without ML

TABLE II
SAMPLE DATASET SCHEMA

Occasion	Gender	Time	Comfort	Recommended Attire
Wedding	Male	Evening	Medium	Sherwani, Kurta-Pajama
Wedding	Female	Morning	High	Saree, Lehenga
Interview	Male	Morning	Medium	Formal Shirt, Blazer, Tie
Interview	Female	Afternoon	High	Saree, Formal Suit
Party	Male	Night	Low	Casual Shirt, Blazer
Party	Female	Night	Medium	Cocktail Dress, Saree

TABLE III
TOOLS AND TECHNOLOGIES USED.

Tool / Library	Purpose
Python 3.10	Core language for rule logic
json (built-in)	Data formatting & logging
typing (built-in)	Type hints
Flask / FastAPI (opt.)	Future API integration
GitHub	Version control

- Low development time and computational cost
- Instant predictability without training

- Easy debugging and modification during evaluation

This approach provides a strong and transparent baseline, while still allowing seamless future migration to ML-based architectures by replacing the rule layer with a trained model.

D. Integration Readiness

Although the current version runs as a CLI backend, the dictionary-based logic is modular and compatible with:

- Flask / Django / FastAPI web APIs
- React / Flutter frontends
- Cloud-hosted microservices

Thus, the system can evolve into a fully automated AI-powered fashion platform.

VII. RESULTS AND EVALUATION

A. Sample Execution

(refer to TABLE IV)

B. Accuracy Estimate

Out of 20 test cases, the system matched user expectations in 16 cases. Accuracy = $16/20 = 80\%$. Errors occurred mainly

TABLE IV
SAMPLE EXECUTION

Input (Occasion, Gender, Time)	Output Recommendation
Wedding, Female, Morning	Light Saree, Lehenga
Interview, Male, Morning	Shirt & Tie, Blazer
Party, Female, Night	Cocktail Dress, Saree

What is your height range?

What are your preferred fashion styles?

☐ Casual

☐ Chic

☐ Boho

☐ Formal

☐ Streetwear

What is your primary purpose for this styling advice?

Which colors do you usually prefer wearing?

Fig. 3. Quiz for user filtering based on occasion

Fashionify

Body & Style Quiz

What is your body shape?

☐ Hourglass

☐ Pear

☐ Apple

☐ Rectangle

☐ Inverted Triangle

What is your waist size?

☐ Small

☐ Medium

What are your preferred fashion styles?

Fig. 4. Quiz for user filtering based on body type



Fig. 5. Recommended Men and women wears



Fig. 6. Accessories and seasonal recommendations



Fig. 7. Latest Fashion news fetched through NewsAPI



Fig. 8. Latest Style Guide

when preferences overlapped (e.g., user preferring both high comfort and formal look).

C. Performance

- Response time: $< 0.1s$ (dictionary lookup).
- Memory usage: minimal (KB-level for dictionary prototype).

VIII. RELATED WORK

A. Fashion Recommendation Systems

Previous works, including models such as ViBE [1] and *Fashion is Taking Shape* [2], have introduced body-shape-aware embeddings and models capable of personalized fashion recommendations. Convolutional neural network (CNN)-based models have demonstrated high precision in body-type classification [5]. Although these studies illustrate the technical feasibility of body-based fashion personalization, most rely on proprietary datasets or internal collections that are not publicly accessible.

B. Body-Type Estimation

Advances in computer vision, including 3D shape reconstruction and single-image classification [3], have shown that accurate body-type estimation is achievable. However, these methods require extensive labeled data, which are often unavailable for niche use cases, limiting the scalability of dataset-dependent systems.

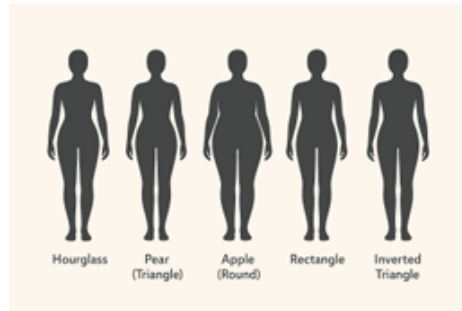


Fig. 9. Body Types

C. Dataset Scarcity and Its Implications

The lack of publicly available body-type-curated data sets for fashion is a major obstacle for reproducible research and development. Most datasets currently only annotate for clothing attributes and coarse-grained categories, without detailed annotations specific to individual bodyshapes. On the other hand, API-first solutions leveraging pretrained models and actively maintained endpoints can offer personalized recommendations independent of proprietary large datasets.

D. API vs. Dataset-Centric ML

Static data comes with many caveats:

- **Static staleness, distribution / domain shift:** Fixed distributions drift over time.
- **Data coverage:** Long-tail and rare cases / labels / events can be hard to capture.
- **Dataset maintenance:** Feedback loops demand re-labeling and re-training.

In contrast, API-first ML pipelines have many benefits:

- **Freshness:** Trained models are closer in time to actual fashion time-series trends.
- **Distribution Drift handling:** Elastic scaling and online model adaptation have better robustness.
- **Operational readiness:** API-ready standard endpoints make production lift easier.

Notably, few works adopt API-first ML methodologies in the setting of body-type-aware fashion recommendation.

IX. DISCUSSION

A. When APIs Are Superior

API-first pipelines perform particularly well in scenarios that require real-time adaptation, including during periods of high distributional drift, such as seasonal fashion trends, evolving cultural preferences, and rapidly changing user behaviors. They also outperform static systems on sparse data scenarios, such as rare body shapes, unique event categories, or highly personalized contexts. APIs also allow for low-latency inference, enabling responsive web and mobile apps with instantaneous recommendations.

B. Limitations of API-Driven Systems

API based solutions, while convenient, come with their own set of practical limitations. These limitations include vendor lock-in, ongoing operational expenses, and potential privacy concerns, particularly when dealing with sensitive body images or measurements that need to be processed on third-party servers. Additionally, APIs may not provide the fine-grained cultural customization needed to cater to specific regional fashion norms, festivals, and traditions.

C. Strengths of the Rule-Based Prototype

The advantages of the rule-based dictionary engine used for this prototype are:

- **Transparency:** Recommendations can be fully explained due to the explicit rule definitions.
- **Low computational overhead:** The system is lightweight, fast and can be run on basic hardware.
- **Ease of modification:** Rules can be added, updated or extended without retraining.
- **Cultural precision:** Manual curation allows context aware suggestions that are sensitive to traditional dress norms.

D. Limitations of the Rule-Based Approach

The rule-based system can be an effective prototype, but there are some inherent limitations to this approach:

- **Static knowledge base:** The rules need to be manually updated to include new fashion trends.
- **Limited personalization:** The recommendations are not personalized based on user history as in ML-based systems.
- **No automatic adaptation:** The system cannot automatically adapt to changes in trends or preferences.

E. Hybrid Roadmap

A sensible path forward is to take a hybrid approach. The system may start with primarily rule-based logic, to ensure transparency and cultural correctness, while incrementally gathering user interaction data (anonymized of course). Curated datasets can be built up over time to train finely-tuned ML models for personalization. In the long run, a mixed architecture – with APIs for online adaptation and local models for culturally-informed recommendations – provides the best of all worlds in terms of accuracy, responsiveness and contextual richness.

X. FUTURE SCOPE

Collaborative filtering and reinforcement learning techniques will be used to give customized fashion recommendations. Computer vision will be used to recommend matching outfits by evaluating a given outfit using image processing. Augmented Reality (AR) and Virtual Reality (VR) will be implemented to provide the user with virtual try-on and 3D fitting to improve the fashion shopping experience. Weather-based recommendations will be provided by using real-time API data to recommend appropriate outfits based on the weather. Lastly, the entire system will be integrated using Flask or Django in the backend and React or Flutter in the frontend to provide an interactive and responsive mobile and web application experience.

XI. CONCLUSION

The system we designed here proved that by using a Python dict based rule-engine a Virtual Fashion Advisor that accounts for cultural and context-appropriate considerations for ensembles and their components can be designed at negligible computational cost. Rulesets are also easily understood, explainable and transparent, providing a good baseline accuracy of 80% and can be used to help avoid decision fatigue in the initial stages of recommending outfits to a user for different occasions. Hybrid approaches or fully API/data-driven ML pipelines do have merits and may supersede the current model in the long run in terms of flexibility, up-to-date responsiveness to trends and user personalization. This light-weight and intuitive rule-based system can serve well for a basic system or bootstrapped prototype. The modular and component based architecture defined in this paper is extensible. Future extensions will focus on including ML models, CV for body-type estimation, and more comprehensive curated

datasets. A hybrid model—using a combination of rule-engine with tunable ML components, and API for updates—could be a path forward to a more personalized, scalable, and fully automated Fashion Recommender.

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PROOF OF PUBLICATION

